

Physics of the Skeleton

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The skeleton is one of the largest organs in the human body and it constitutes up to approximately 15% of total body weight, Bones serve various functions that affect the whole body

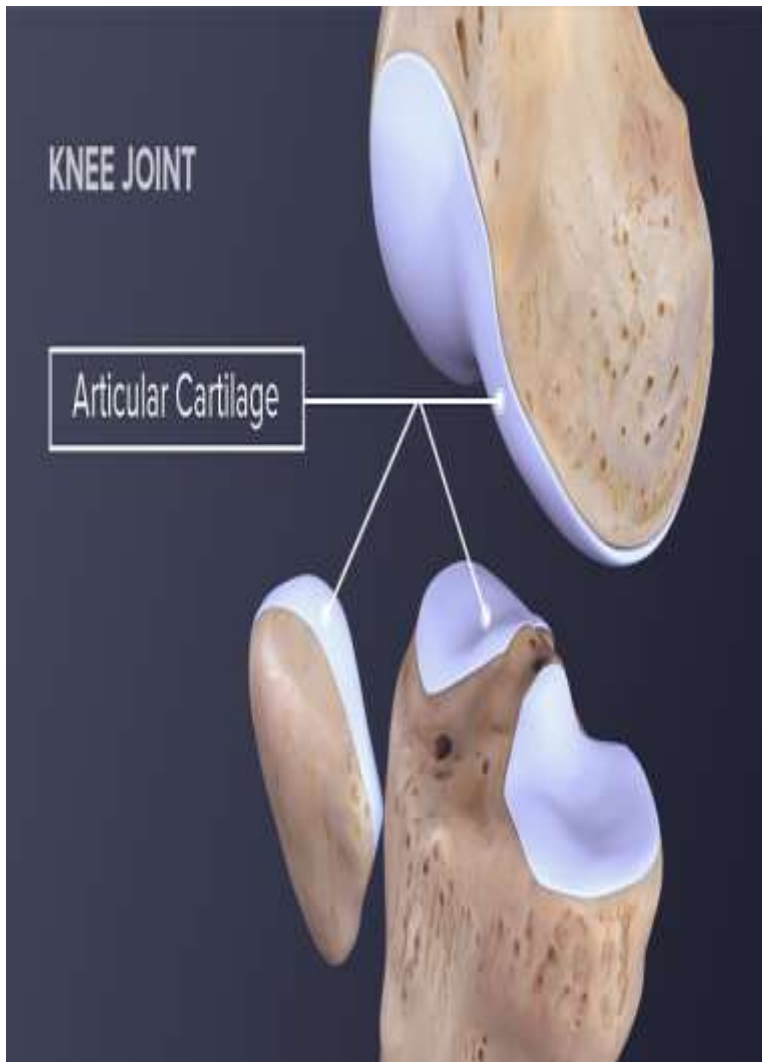
1- Mechanics

Bones provide a frame to support the body. Muscles, tendons, and ligaments attach to bones. Without anchoring to bones, muscles could not move the body.

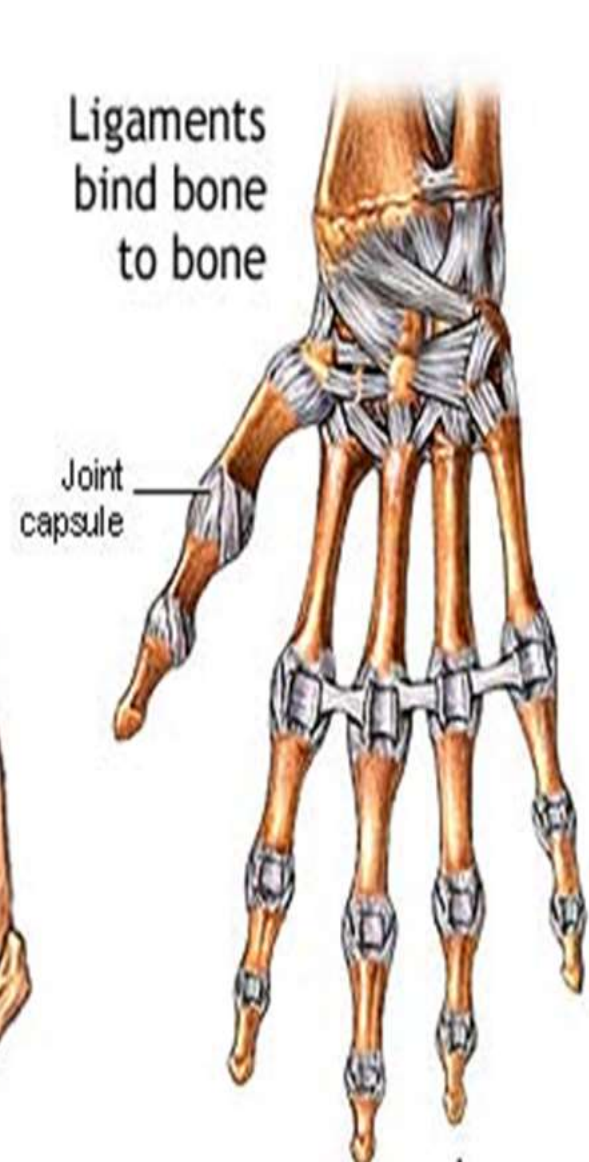
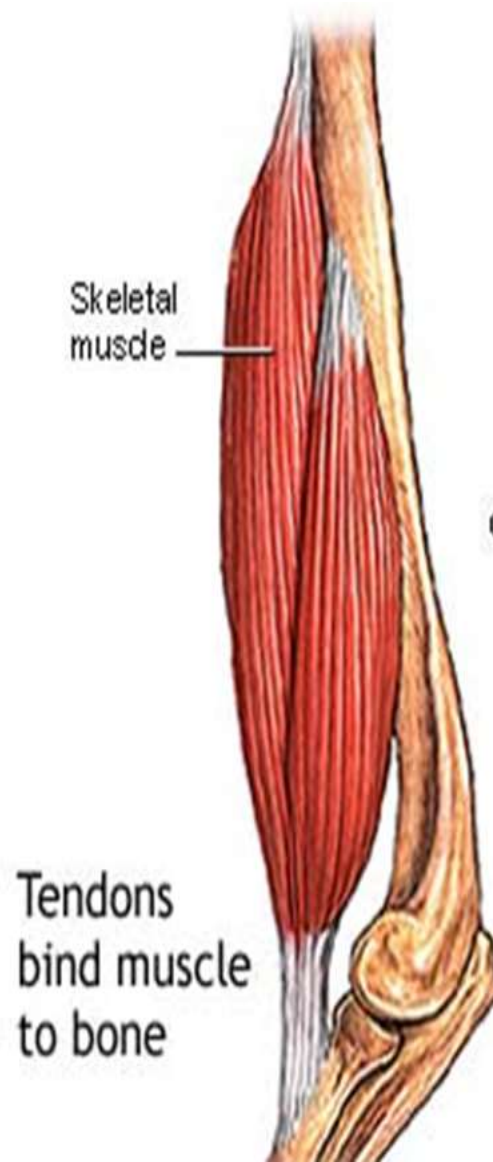
The system of bones and muscles support the body.

- **Ligaments**
- **Tendons**
- **Cartilage**





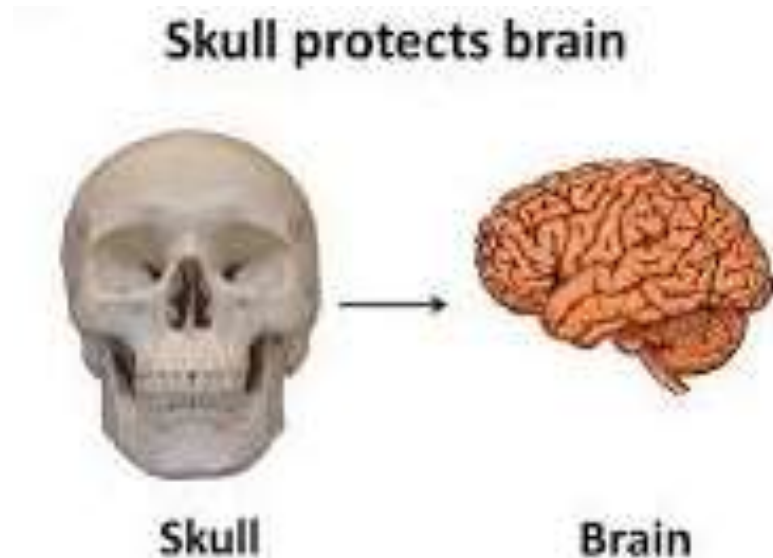
Cartilage at the end of bones reduce friction and prevents them from rubbing together when their movement



- **Tendons bind muscle to bone**
- **Ligament bind bone to bone**

2- Protection

Some bones protect the body's internal organs. For instance, the skull protects the brain, and the ribs protect the heart and lungs.



3- Stores minerals – calcium

Storage chemicals: -

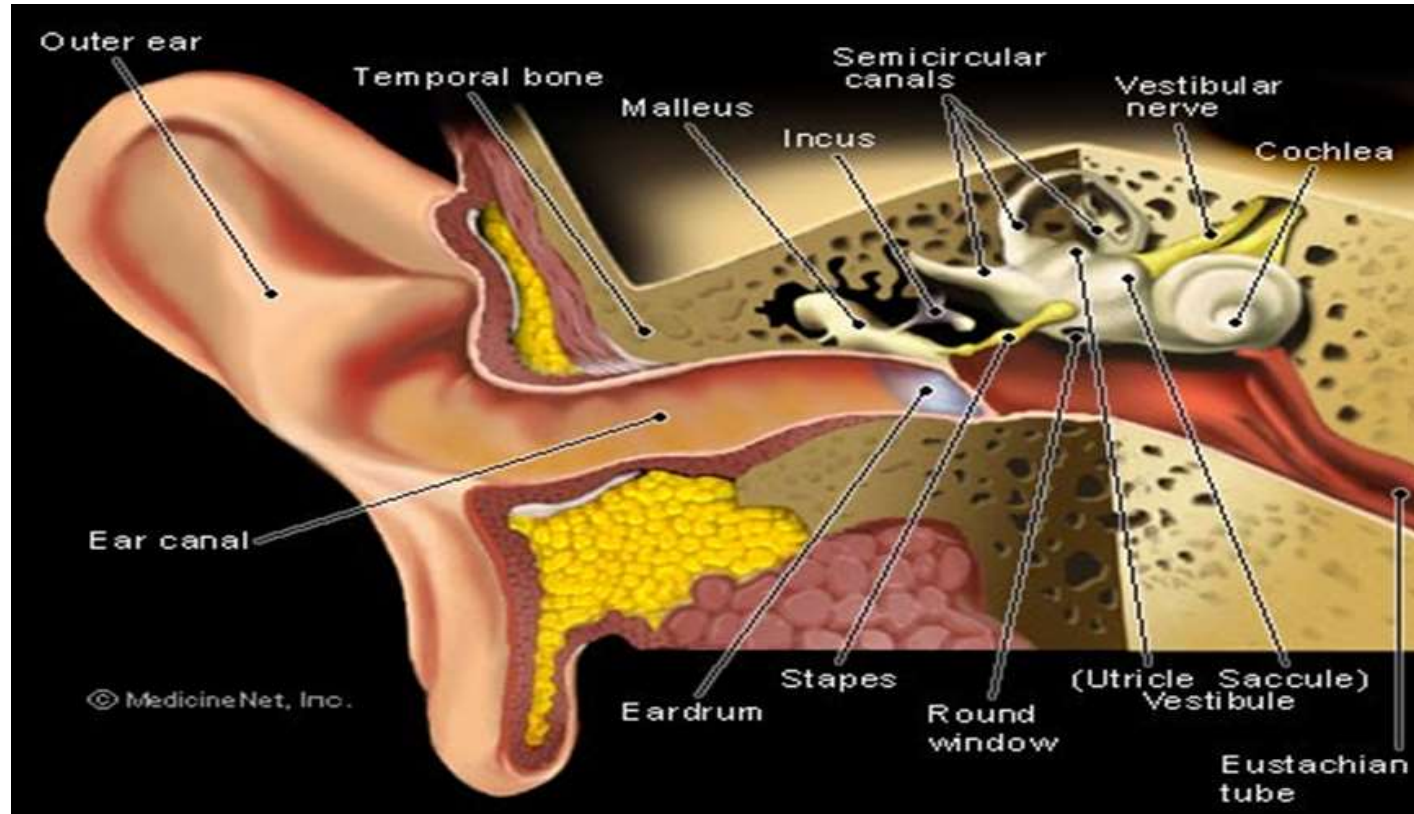
Bones act as a bank or storing element for future in the body.

Example: -

A minimum of calcium is needed in the blood, if the level falls too low, the Ca^{+2} sensors causes the parathyroid glands to release more parathyroid hormones into the blood. This causes the bones to release the needed Ca^{+2} .

4- Allows for body movement

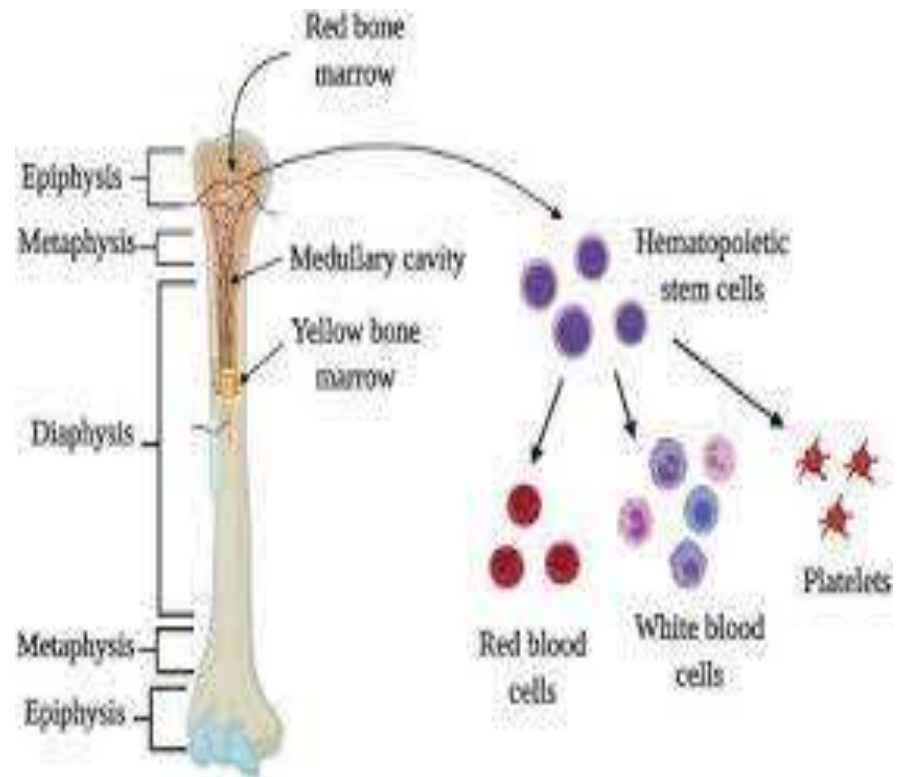
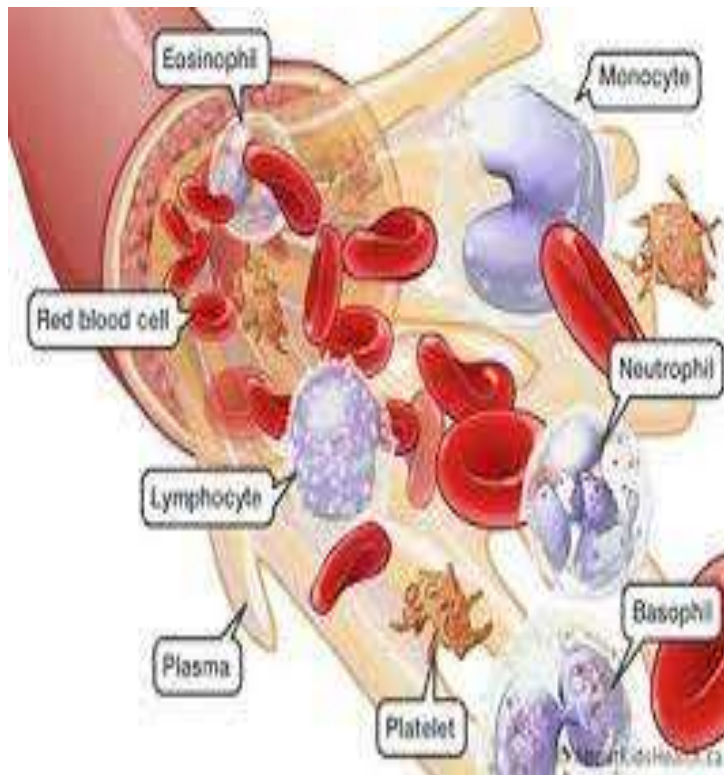
5- Sound transmission in the middle ear.



The smallest bones of the body are the "ossicles" in the middle ear, they act as an impedance matching system for converting sound vibration in air to sound vibration in fluid in the "cochlea". They attain full adult size before birth.

6- Synthesis

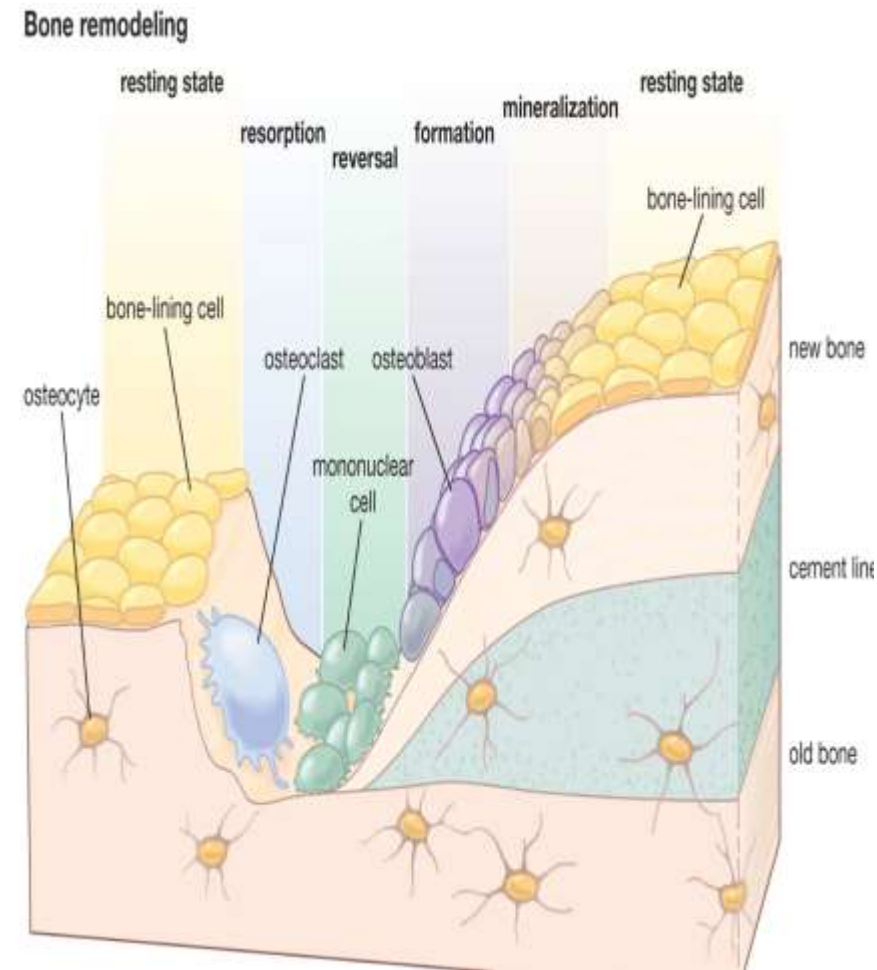
Cancellous bone is a vital reservoir for developing red blood cells, platelets, and white blood cells. Also, the body destroys defective and old red blood cells in bone marrow.



Bone is composed of four different cell types; osteoblasts, osteocytes, osteoclasts and bone lining cells.

Osteocytes

Osteocytes are cells inside the bone. They also come from osteoblasts. Some of the osteoblasts turn into osteocytes while the new bone is being formed, and the osteocytes then get surrounded by new bone. Cells that maintain the bones in a healthy condition, they make about 2% of the volume of the bone. If these cells die, the bones die (due to a poor blood supply).



Bone remodeling:

A process performed by specialized bone cells during human life period. These cells are called: -

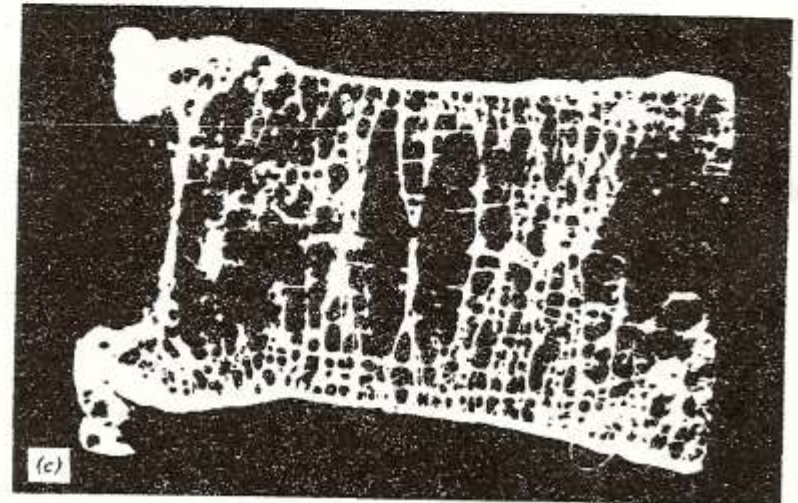
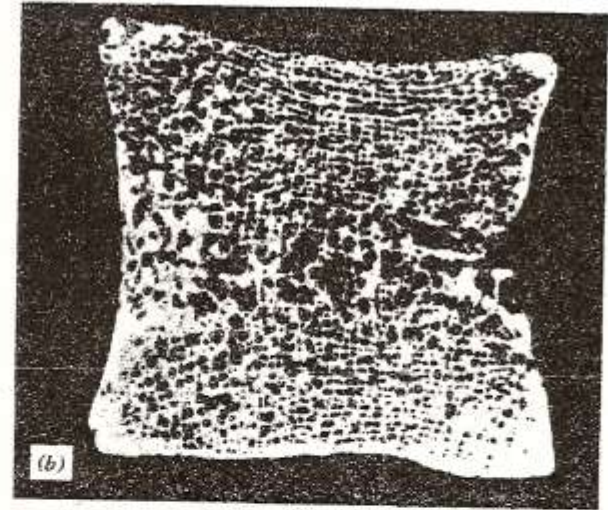
1. **Osteoblasts** which build up bones using Ca^{+2}
2. **Osteoclasts** which destroy the bone.

Up to 35-40 years, the activity of the osteoblasts is greater than the osteoclasts.

After this age, osteoclasts is more active resulting in general decrease in the bone mass, that continues until death. This is called "osteoporosis" or (porus bones).

Osteoporosis or (porous bones).

Osteoporosis is a bone disease that involves a reduction in bone mineral density. This increases the risk of fractures.



a: Normal and healthy bone, b: Osteoporosis

What bones made of?

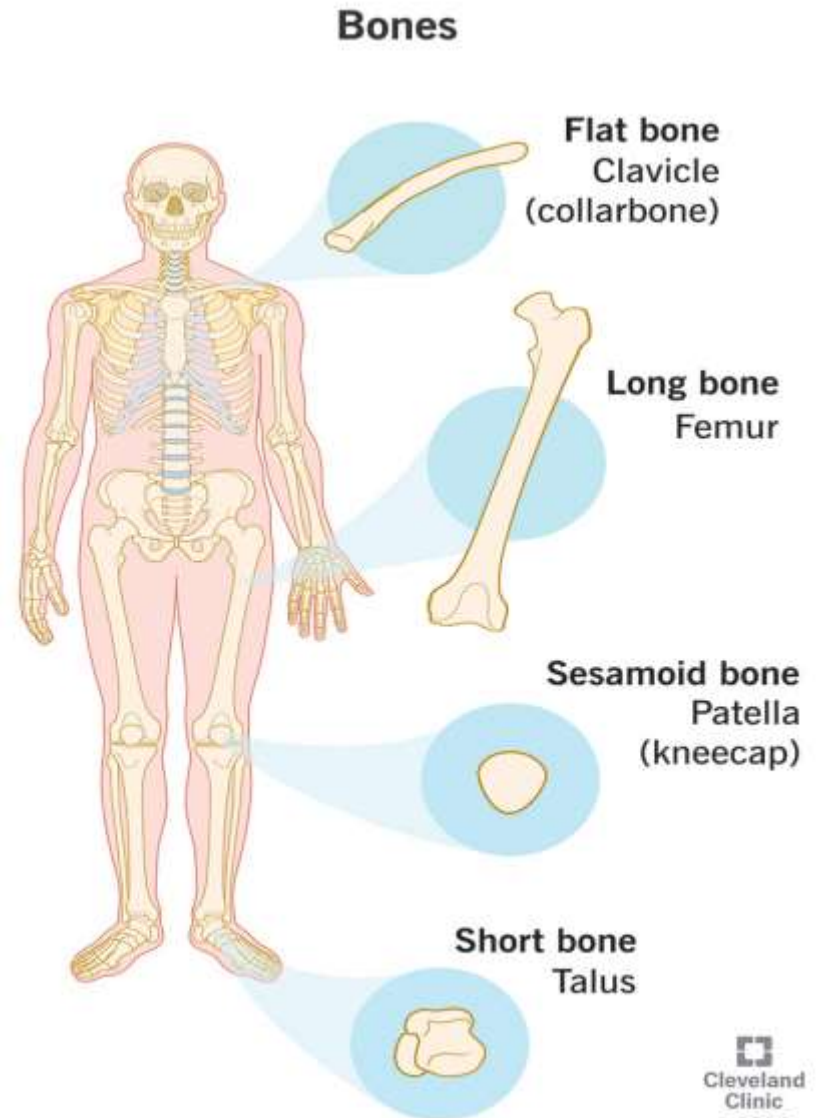
Bone consists of two different materials plus water:-

a. Collagen: about 40% of the weight of the solid bone and 60% of volume.

b. Bone minerals (inorganic): about 60% of the weight of the solid bone and 40% of volume.

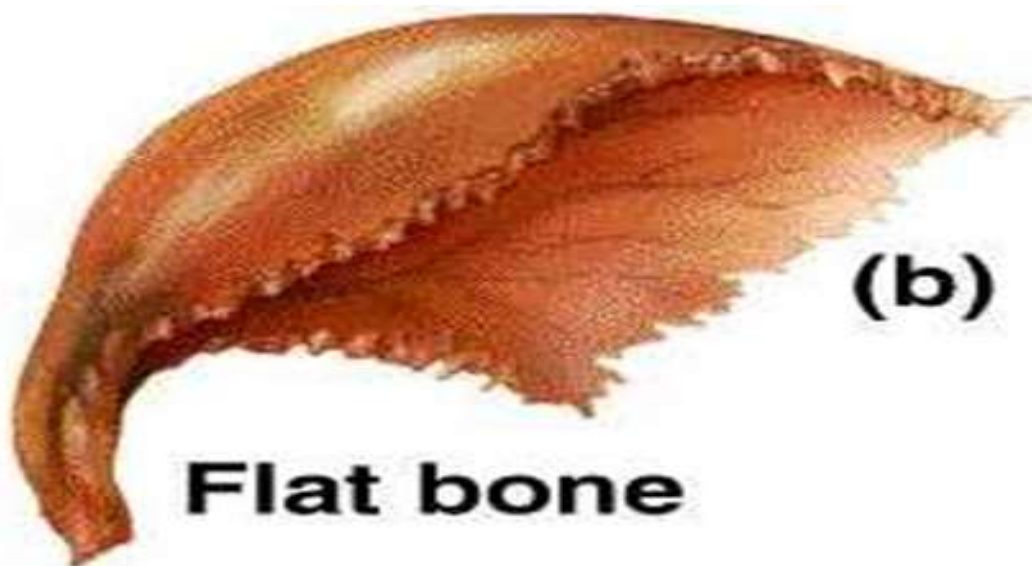
Types of Bones (the 206 bones in human body)

- 1-Small pile: flat, plate like bone as scapula and some bones of the skull.
2. Long hollow bones: - in arms, legs and fingers.
3. Irregular bones: - wrist and ankle
4. More or less cylindrical bone from spine (vertebra).
5. Ribs.
6. A sesamoid bone is a small bone commonly found embedded within a muscle or tendon near joint surfaces





Long bone



Flat bone



Irregular bone



Short bone

What is movable bone in the skull?

- The mandible is the only movable bone in the skull.
- It is the strongest bone in **the face** . Mandible is not the strongest bone in our body. The strongest bone in the body is **the femur** which is located in the thigh

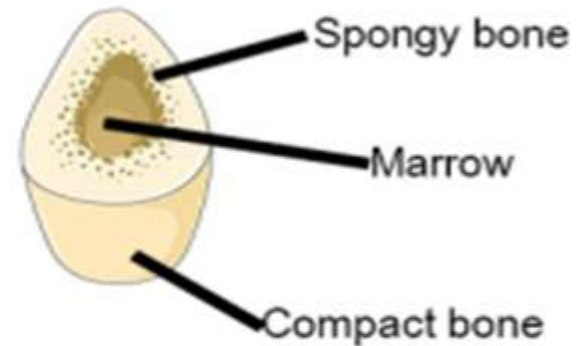
Choose True or False

Is the Mandible the strongest bone in the face? **True**

Is the Mandible the strongest bone in the body? **False**

Is the femur the strongest bone in the body? **True**

Bone Structure



1. Solid or compact bone.

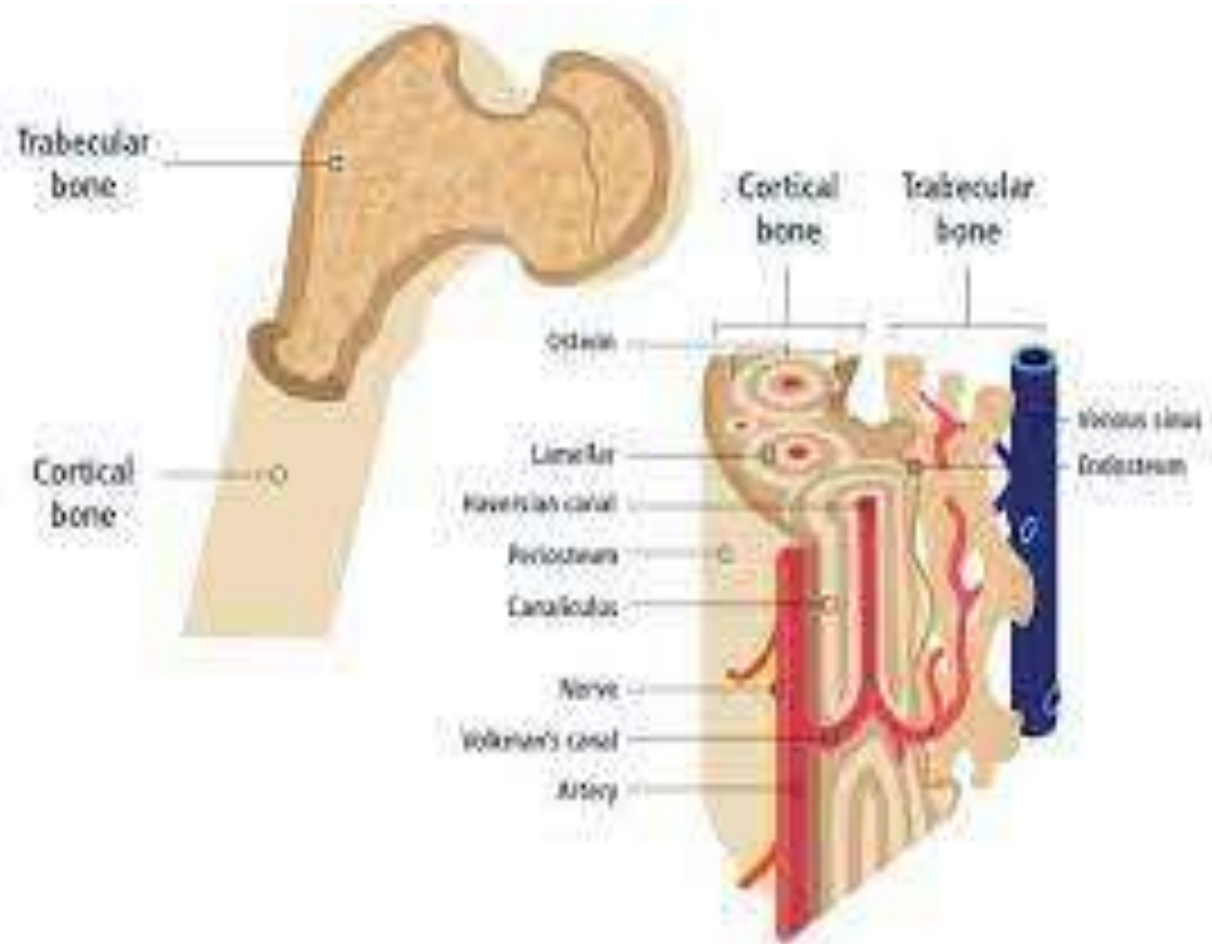
2. Spongy or cancellous (thread like trabecular bone).

3. The medullary cavity is the hollow part of bone that contains bone marrow. The bone marrow makes blood cells and stores fat

so can find bones composed of one or a combination of two different types of bones: -

Example: -

In long bones, the long shaft is compact; while the ends are spongy (ends are weaker deal to the amount of bone (volume)).



What are the advantages of trabecular bone over compact bone?

- 1. Where a bone is subjected primarily to compressive forces (such as the ends of the bones), trabecular bone gives the necessary strength with less material than compact bone.**
- 2. The trabecular are relatively flexible, they can absorb more energy when a large forces are involved such as in walking, running and jumping.**

- On the other hand trabecular bone cannot withstand well the bonding stress that occurs mostly in the central portions of the long bone.

Bone density

The density of compact bone is constant $\approx 1.9 \text{ gm/cm}^3$ or (1.9 times the density of water).

Young's modulus for a bone

When a sample of fresh bone is placed under force then:

Stress is a quantity that describes the magnitude of forces that cause deformation. Stress is generally defined as force per unit area. When forces pull on an object and cause its elongation

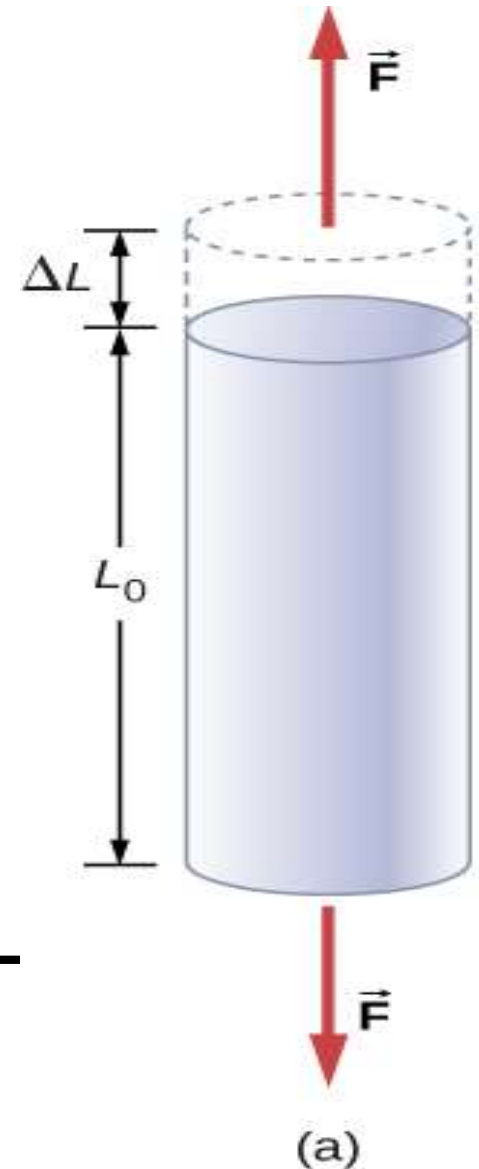
$F/A = \text{stress} \implies \text{Hook's law}$

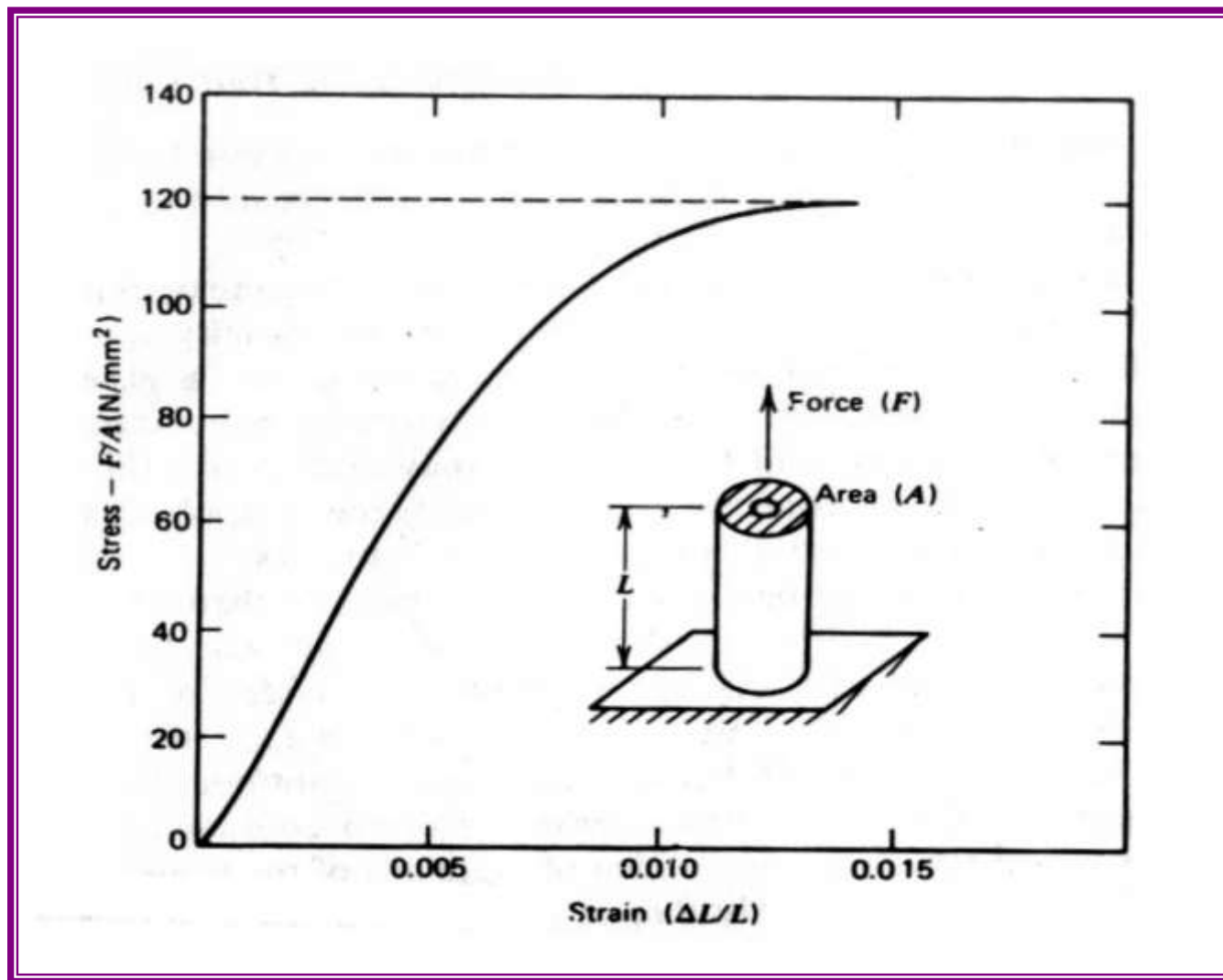
The sample under such force will increase in length ΔL ;

$\Delta L/L = \text{strain}$

The ratio of stress to strain ($F \cdot L / A \cdot \Delta L$)

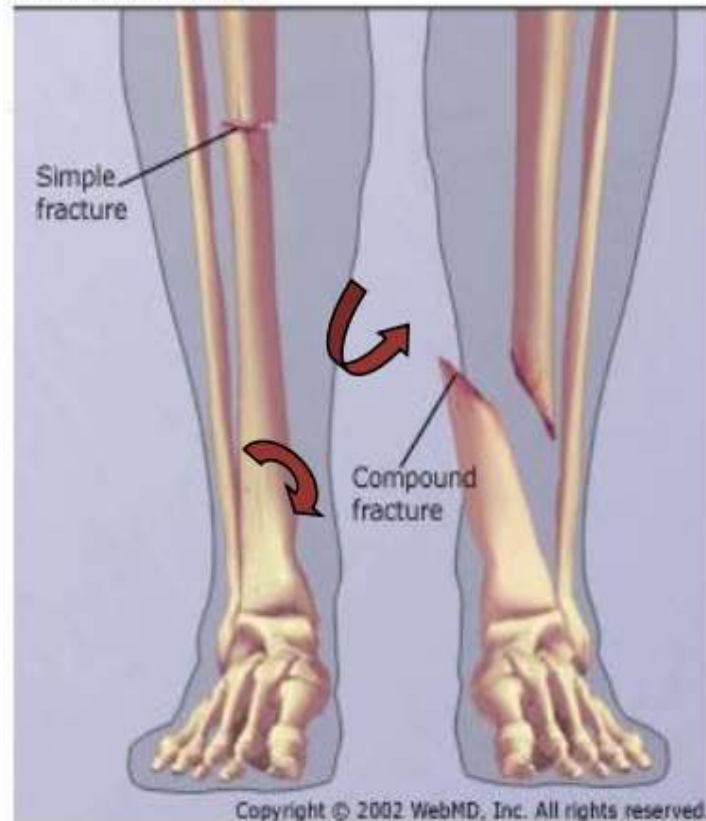
Young's Modulus Y





the tensile breaking stress is 120 (N/mm^2)

Bone Fractures



The bones don't normally break due to compression; they usually break due to shear or under tension. A common cause of shear is catching the foot and then twisting the leg while bone is apt to puncture the skin, this type of fracture is compound.

Measurements of bone mineral in the body

The strength of bones depends on the mass of bone mineral present.

Bone mineral mass decreases slowly 1-2% per year, so very precise technique is needed.

Measurement of Bone Density:

A bone mineral density (BMD) test can provide a snapshot of your bone health. The test can identify osteoporosis, determine your risk for fractures (broken bones), and measure your response to osteoporosis treatment. The most commonly used BMD test is called a central **dual-energy x-ray absorptiometry**, or central **DXA** test. It is painless – a bit like having an x-ray. The test can measure bone density at your hip and lumbar spine.

A BMD test measures your bone mineral density and compares it to that of an established norm or standard to give you a score. Although no bone density test is 100-percent accurate, the BMD test is an important predictor of whether a person will have a fracture in the future.



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Dual-Energy x-ray Absorptiometry (DXA) test

III/ In vivo activation

Whole body is irradiated with energetic neutrons this converts Ca^{+2} and some other elements into radioactive form. This detected to give the amount of Ca^{+2} .

BM is obtained by multiplying a constant.

Bone mineral mass (BM gm/cm²) = $K \log I_0/I$.

Disadvantages

- 1. Expensive technique.**
- 2. Hazard of large radiation exposure.**

Short Answer Questions

1. Mention three functions of the skeletal system.
2. Define bone remodeling.
3. Name the four types of bone cells.
4. What happens when blood calcium level falls below normal?
5. List two advantages of trabecular bone over compact bone.
6. What is osteoporosis?
7. What are the two main solid components of bone?
8. Which bones protect the brain and the heart?
9. What is the function of the medullary cavity?
10. Define stress and strain in bone mechanics.

Essay / Long Answer Questions

1. Describe the mechanical function of bones and the role of tendons, ligaments, and cartilage.
2. Explain bone remodeling, including the role of osteoblasts and osteoclasts and age-related changes.
3. Discuss calcium storage in bone and its regulation in the body.
4. Describe the structure of a long bone, including compact bone, spongy bone, and medullary cavity.
5. Explain the in vivo activation method for measuring bone mineral mass, including its disadvantages.

Thank you